Homework 16

For Monday, November 16, 2026.

Give exact answers and justify your reasoning. Do not use electronic tools. Angles are measured in radians.

1. A semicircle of radius $R > 0$ above the x-axis has equation $y = \sqrt{R^2 - x^2}$. Rotate the region under the semicircle about its diameter. Derive the volume of the resulting ball by integrating disk areas.

2. Rotate the region $0 \le x \le 1$, $0 \le y \le x^2$, about $y = 2$. At $x = 1/2$, give the inner and outer radii of the washer. Then find the volume of the solid.

3. A region to the left of the line $x = 3$ is rotated about that line. Each horizontal slice of the region is one segment. Its washer integral is

$$\pi \int_0^1 \left[(3 - y^2)^2 - (3 - y)^2\right] dy.$$

Recover the endpoints of a horizontal slice as functions of y. Sketch the region and the axis of rotation.

4. Rotate the parallelogram $0 \le x \le 2$, $-x \le y \le 1 - x$, about the x-axis. At which values of x does the formula for the cross-sectional area change? Write a piecewise integral for the volume and evaluate it.

5. A solid of revolution has disk sections with radius $r(x)$ decreasing along its axis. Measurements in centimetres give $r(0) = 3$, $r(1) = 2$, and $r(3) = 1$. Use just these data and monotonicity to give a lower and an upper bound on the volume between $x = 0$ and $x = 3$.

6. Rotate the region $0 \le x \le 4$, $0 \le y \le \sqrt{x}$, about the x-axis. Find the plane $x = c$ that divides the resulting solid into two equal volumes.

7. A cylindrical shell has height 3, inner radius 4 and outer radius $4+h$, where $h>0$. Find its exact volume $V(h)$. If the volume is estimated as inner circumference times height times thickness, by how much is the estimate too small? Find $\lim_{h\to 0^+} V(h)/h$.

8. Use horizontal shells to find the volume obtained by rotating the region $0 \le y \le 1$, $0 \le x \le e^y$, about $y=-1$.

9. Rotate the rectangle $-1 \le x \le 2$, $0 \le y \le 3$, about the y-axis. The expression $2\pi \int_{-1}^{2} 3|x|\,dx$ counts part of the solid twice. Identify the radii counted twice, find the excess volume and give the actual volume.

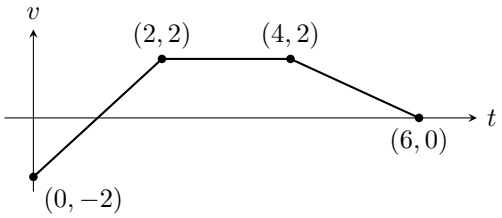
10. The region under $y = x(1-x)^2$ on $[0,1]$ is rotated about the y-axis. Choose a method that lets you compute the volume without solving a cubic equation. Explain the choice and evaluate the integral.

11. Rotate the region $1 \le x \le 4$, $1 \le y \le \sqrt{x}$, about the y-axis. Set up the volume once using washers and once using shells, labeling the radii and shell height. Evaluate either integral.

12. Let f be continuous and nonnegative on $[a,b]$, with $0 \le a < b$. The region under f has area A. Let V_0 be its volume of revolution about the y-axis and V_d its volume of revolution about $x=-d$, where $d>0$. Use shells to express $V_d - V_0$ in terms of A and d.

13. Find the average value of $f(x) = 1/x$ on $[1, 4]$. At what point in the interval does f take that value?

14. The graph shows the velocity $v(t)$ of a particle in metres per second, with t in seconds. Consecutive marked points are joined by line segments. Find the particle's average velocity and its average speed on $[0, 6]$.



15. An integrable function has average value 5 on $[1, 7]$ and average value 8 on $[1, 3]$. Determine its average value on $[3, 7]$.

16. Interpret the following limit as the average value of $\sqrt{x}$ on a suitable interval. Use that interpretation to find its exact value.

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \sqrt{4 + 5k/n}.$$

17. A temperature model is $T(t) = 18 + 4\cos(\pi t/12)$ degrees Celsius, with t in hours. Show that the average temperature on $[s, s + 24]$ is the same for every starting time s, and find that average.

18. Let f be continuous on $[0, b]$. For $0 < t < b$, let $M(t)$ be its average value on $[0, t]$. Express $M'(t)$ in terms of $f(t)$, $M(t)$ and t. When is $M'(t) > 0$?

19. A tank extends along an axis measured in metres. The volume from $x = 0$ to $x = t$ is $V(t) = \pi(4t + t^2)$ cubic metres for $0 \le t \le 3$. Find the average cross-sectional area between $x = 1$ and $x = 3$. What radius would a circular cylinder of the same length need to hold this portion of the tank's volume?

20. Let f be continuous on $[a, b]$, where $a < b$. Apply the Mean Value Theorem to $F(x) = \int_a^x f(t)\,dt$ to prove that $f(c) = f_{\text{avg}}$ for some $a < c < b$. Give an example of an integrable function with a jump whose average value is never attained.